

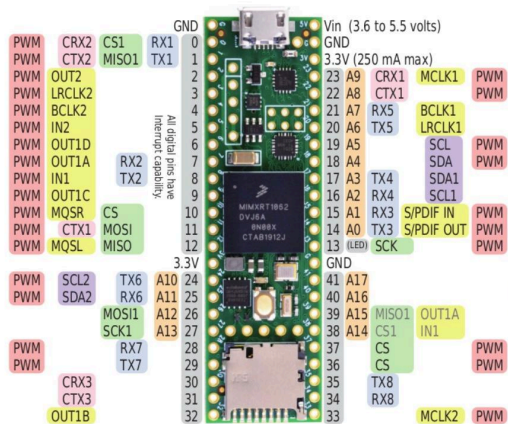
Hardware selection

- **Teensy 4.1**

Pros: very fast, many hardware serial/I/O ports, and a **built-in SD card slot**, which is useful for reliable logging.

Cons: no built-in Bluetooth or Wi-Fi, so extra modules are needed.

Comparison: compared with an **ESP32**, the ESP32 has built-in wireless, but the Teensy is better for this project because it offers more processing headroom, more separate ports, and easier SD-based logging.



- **Bluefruit LE UART Friend**

Pros: easy to use, reliable BLE communication, and works well with the **Adafruit app**, so no custom phone app was needed.

Cons: adds an extra module and extra wiring instead of having Bluetooth integrated on the microcontroller.

Comparison: compared with a generic Bluetooth module, this one was more practical because of the ready-to-use app support.

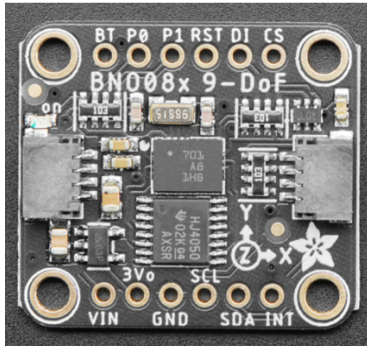


- **BNO085 IMU**

Pros: provides fused orientation data directly, so roll, pitch, and linear acceleration are easier to measure accurately.

Cons: more expensive than a basic raw IMU.

Comparison: compared with a simple accelerometer/gyroscope module, the BNO085 reduces software complexity because less sensor-fusion code is needed.



- **PA1616D GPS**

Pros: supports GPS + GLONASS, higher update rates, and connection to an external antenna.

Cons: GPS performance still depends on satellite visibility and antenna placement.

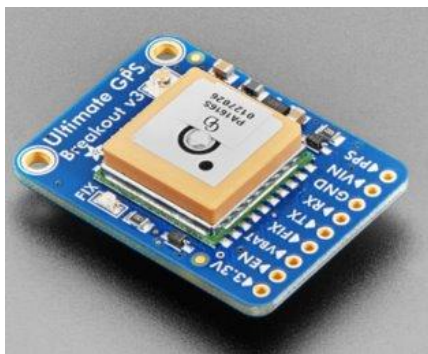
Comparison: compared with simpler older GPS modules, it is more suitable for motion tracking and gives more robust positioning.

- **External active antenna**

Pros: improves GPS reception.

Cons: adds size and another external component.

Comparison: compared with using only the onboard GPS antenna, it gives better performance when the system is mounted in an enclosure.



Conclusion:

Overall, these components were chosen because they give a good balance between performance, ease of use, and reliable logging on a custom PCB.

PCB

- **Teensy interface**

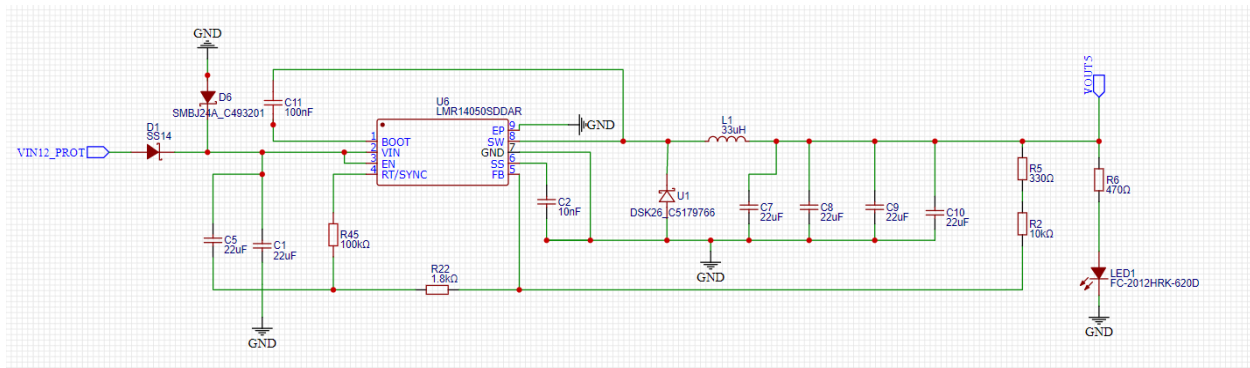
The PCB is built as a carrier/interface board for the Teensy 4.1 using the two side header rows. These headers bring out the required UART, I2C, analog, interrupt, and GPIO signals to the external modules and sensors. The routing is split over the left and right header banks so the external wiring stays organized.

- **12 V input protection**

The board is powered from the motorcycle's switched 12 V input. At the entrance of the PCB, D1 is placed in series to block reverse polarity or accidental backfeed, and D6 is connected as a TVS clamp to suppress voltage spikes from the motorcycle electrical system. This makes the input stage more robust for automotive use.

- **5 V power conversion**

The protected 12 V line is converted to 5 V by the buck converter stage. The power section includes the switching IC, freewheel diode, inductor, input/output capacitors, and a 5 V indicator LED. Functionally, this creates the main supply rail VOUT5 for the GPS, BLE module, brake sensor supply, and other external interfaces. Reusing the previously tested converter reduced risk and kept the supply section predictable.



- **Hall sensor conditioning**

The two external motorcycle hall sensor inputs are processed on the PCB before reaching the Teensy. Since the measured signal levels were only about 0.6 V to 1.7 V, they were not suitable for direct digital input.

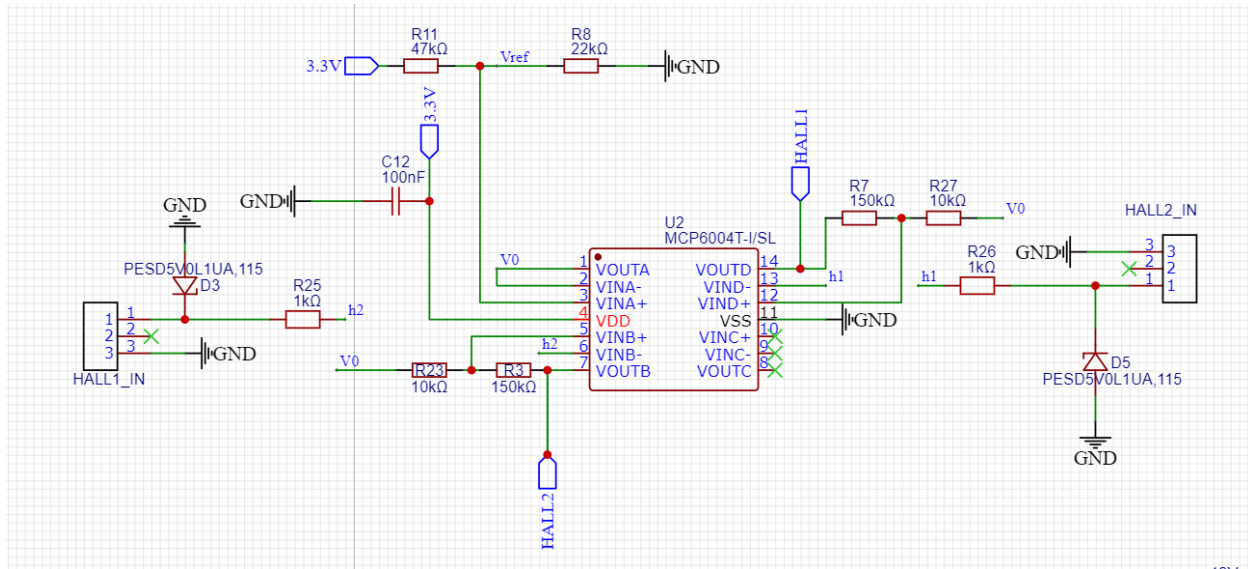
The front-end uses the MCP6004 as two Schmitt-trigger comparators with hysteresis:

- Vref is generated by 47 kΩ / 22 kΩ, so
$$V_{ref} = 3.3 \times 22 / (47 + 22) \approx 1.05 \text{ V}$$
- With 150 kΩ positive feedback and 10 kΩ reference path, the switching thresholds are about
0.99 V and 1.19 V
- So the hysteresis is about
$$1.19 - 0.99 = 0.20 \text{ V}$$

This ensures clean switching and prevents false triggering when the hall signal hovers near the threshold.

- **Hall input protection**

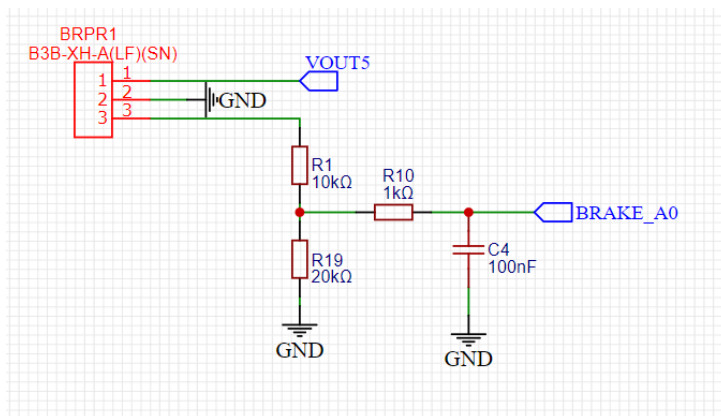
Each hall input also has an ESD protection diode to ground. This protects the analog front-end from transient voltages arriving through the motorcycle harness.



- **Brake pressure input**

The brake pressure connector includes power, ground, and signal. The signal is scaled before the Teensy ADC using 10 k Ω / 20 k Ω :

- $V_{out} = V_{in} \times 20 / (10 + 20) = 0.667 V_{in}$
 - So a 5.0 V sensor output becomes about 3.33 V at the ADC input
- After the divider, 1 k Ω + 100 nF forms a simple low-pass filter with $\tau = RC = 1 \text{ k}\Omega \times 100 \text{ nF} = 100 \mu\text{s}$, which helps suppress high-frequency noise.



- **Module connectors**

The IMU, GPS, and Bluetooth modules are connected through female headers, making

the PCB modular and easy to assemble or service. The IMU connector carries 3.3 V, GND, SDA, SCL, and INT. The GPS connector carries 5 V, GND, UART, FIX, and PPS. The BLE connector carries 5 V, GND, and UART.

- **Spare expansion connectors**

DEV1 and DEV2 are routed to unused Teensy pins and kept available for future functions. Electrically, they act as reserved expansion points so the board can be extended without redesigning the main PCB.

- **Layout strategy**

The board is arranged by function:

- power stage near the 12 V input
- hall-conditioning stage near the hall connectors
- sensor/module headers around the Teensy interface

This keeps noisy power switching away from low-level signal conditioning and keeps external routing as short as possible and well structured.

- **Board operation**

The PCB takes the motorcycle's switched 12 V input and generates the main 5 V rail with the buck converter. This 5 V rail powers the Teensy, while 3.3 V is supplied from it to the modules and signals that require it. All external signals are then interfaced through the PCB: the hall sensor signals are conditioned before reaching the Teensy, the brake pressure signal is scaled and filtered for ADC measurement, and the IMU, GPS, and Bluetooth lines are routed through their dedicated headers. Therefore, the PCB functions as the central interface for power conversion, signal conditioning, and signal distribution.

